OSA: Echocardiography Video Segmentation via Orthogonalized State Update and Anatomical Prior-aware Feature Enhancement

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Main Conference

Problem Setting

LV segmentation from echocardiography is essential for cardiac function assessment, yet challenging due to:

- **Severe speckle noise** & low contrast
- **Rapid non-rigid deformation** (systole ↔ diastole)
- **Sparse annotations** (ED/ES frames only)

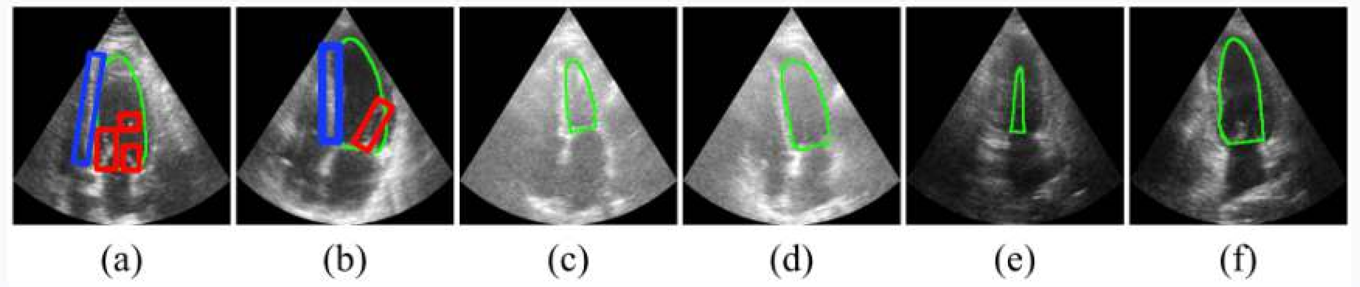

Figure 1. Challenges: speckle noise, blurred contours, and shape variations.

Our Approach

We propose **OSA** — joint temporal stabilization + anatomy-aware enhancement:

- **OSU:** Stiefel manifold constraint prevents rank collapse
- **APFE:** Physics-driven decoupling separates anatomy from speckle

Under sparse supervision (ED/ES only), the objective is:

$$L(\theta) = \ell(\hat{y}_1, y_1) + \ell(\hat{y}_T, y_T)$$

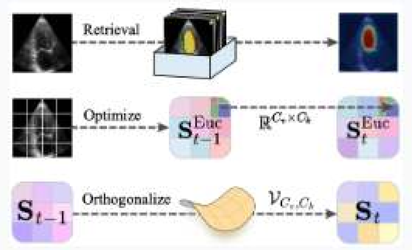

Figure 2. Paradigm comparison: Memory Bank, LRM, and OSA.

Motivation

Existing video segmentation models ignore ultrasound-specific noise statistics, while temporal models suffer from state degradation. OSA addresses both jointly:

Method Overview

OSA segments LV using a **ResNet-50** backbone.

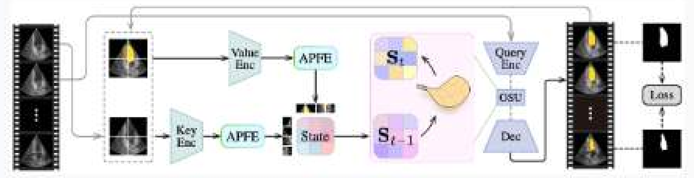

Figure 3. OSA architecture: APFE → OSU → decoder.

Pipeline:

1. ResNet-50 extracts features $\mathbf{F}_t$
2. APFE decomposes into keys/values
3. OSU maintains state $\mathbf{S}_t$ on Stiefel manifold
4. Query $\mathbf{Q}_t \otimes \mathbf{S}_t$
5. Decoder outputs mask $\mathbf{M}_t$

Contributions

- First unified temporal + ultrasound-aware architecture
- Stiefel constraint prevents rank collapse
- Physics-driven decoupling for noise resilience
- **Stiefel manifold:** $\mathbf{S}_t \in \mathbf{V}_{Cv,Ck}$ with $\mathbf{S}^\top\mathbf{S} = \mathbf{I}$
- **Newton-Schulz projection:** avoids exact SVD, O(1) fixed steps

Ablation Study

Full: 94.82 mDice. w/o APFE: 93.61. w/o OSU: 94.12. Both essential.

Method	mDice ↑	mHD95 ↓	Mem ↓	Time ↓
Baseline	92.94	3.56	7.5	4.5
w/o APFE	93.61	3.29	7.6	3.0
w/o OSU	94.12	3.21	7.5	4.7
Full	94.82	3.25	7.6	3.0

Table 2. Ablation study on CAMUS.

$$X_t = X_t^+ - X_t^- + M_t$$

Lossless residual decomposition (APFE)

Orthogonalized State Update

Linear recurrent models suffer **rank collapse**: unconstrained updates decay singular values in $\mathbf{S}_t$.

State Update on Stiefel Manifold

$$S_t = OrthProj_{Stiefel}(S_{t-1} + \eta \cdot \nabla S\ L)$$

via Newton-Schulz iteration + weight decay

OSU projects state updates onto the Stiefel manifold, ensuring stable transitions across the cardiac cycle.

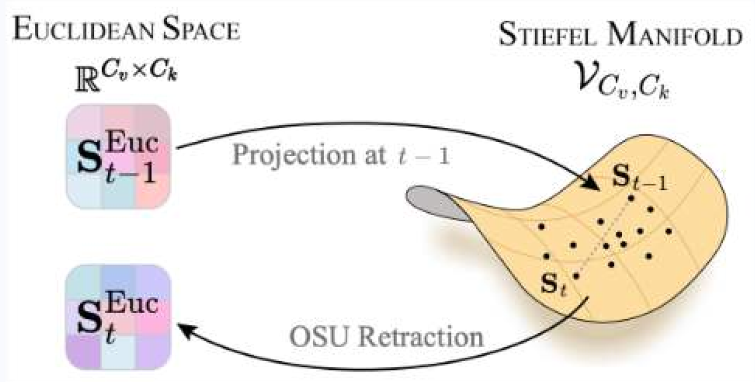

Figure 4. OSU: orthogonalized update preserves manifold structure.

Anatomical Prior-aware Feature Enhancement

APFE decouples anatomy from speckle noise via **physics-driven decomposition**, building stable structural anchors.

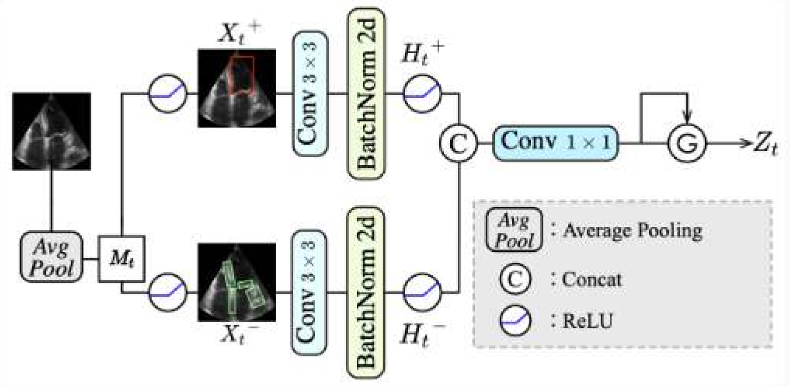

Figure 5. APFE: contrast decomposition provides noise-resilient cues.

Quantitative Results

SOTA on **CAMUS** & **EchoNet-Dynamic** across mDice, mIoU, HD, and ASD.

Method	CAMUS				EchoNet-Dynamic			
	mDice ↑	mHD95 ↓	corr ↑	bias±std (%)	mDice ↑	mHD95 ↓	corr ↑	bias±std (%)
PolaFormer [32] ICLR'25	92.47	3.66	0.792	-0.28 ± 7.5	91.51	3.89	0.678	-0.01 ± 8.1
Vision LSTM [2] ICLR'25	92.71	3.47	0.745	-2.36 ± 7.6	91.71	3.85	0.691	-0.37 ± 7.9
Cutie [9] CVPR'24	91.43	3.97	0.629	0.44 ± 9.8	90.44	4.05	0.600	0.40 ± 9.6
LiVOS [29] CVPR'25	92.89	3.56	0.718	-1.69 ± 7.8	91.92	3.82	0.702	-0.04 ± 7.5
SAMed-2 [50] MICCAI'25	93.55	3.29	0.815	-2.16 ± 6.8	92.67	3.60	0.731	-0.02 ± 7.3
EchoVim [17] ACM MM'25	94.01	3.21	0.804	-0.25 ± 7.2	93.22	3.47	0.803	-0.62 ± 6.2
Vivim [52] TCSVT'25	93.36	3.45	0.817	-0.09 ± 7.3	92.49	3.70	0.758	1.43 ± 7.0
GDKVM [47] ICCV'25	94.18	3.31	0.835	-0.71 ± 6.7	93.33	3.55	0.810	-0.08 ± 6.5
OSA	**94.82**	**3.25**	**0.830**	**-0.44 ± 7.0**	**93.90**	**3.42**	**0.816**	**-1.08 ± 6.0**

Table 1. Segmentation performance on CAMUS and EchoNet-Dynamic.

Dataset Details

CAMUS: 500 patients, A4C + A2C views, 2D+T, ED/ES annotations. Resized to 256×256, 15 frames.

EchoNet-Dynamic: 10,030 A4C videos, 112×112, 10 frames. Sparse labels (ED/ES only).

$$mDice = 2|X \cap Y| / (|X| + |Y|)$$
$$mIoU = |X \cap Y| / |X \cup Y|$$

Clinical metrics: LVEF via Simpson's rule, Pearson **r**, bias±std.

Qualitative Comparison

OSA shows superior boundary adherence and fewer false positives in speckle-dense regions.

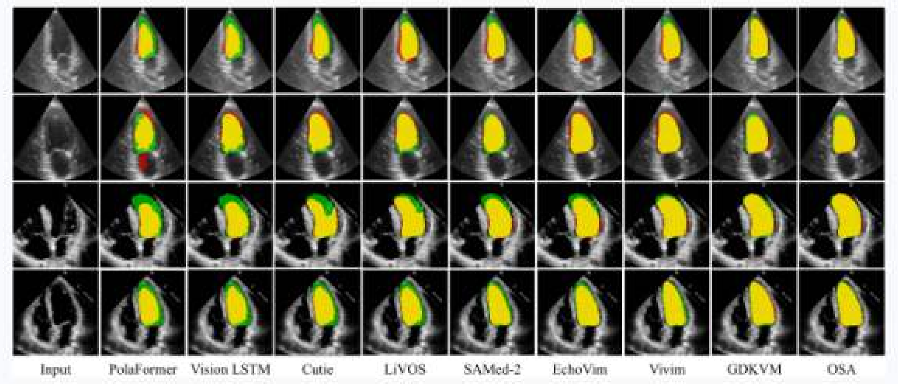

Figure 6. Visual comparison. Green = GT, Red = pred, Yellow = overlap.

Code

github.com/wangrui2025/OSA

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